

Determining Potential Yeast Longevity Genes via Optimized K-Means Clustering

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RESULT

This study employed the Silhouette Score as the definitive quantitative evaluation metric, conducting a continuous parametric sweep across a K range from 2 to 10.

Number of Clusters (K)	Silhouette Score	Cluster Sizes
K = 2	0.5785	[101, 69]
K = 3	0.5283	[74, 56, 40]
K = 4	0.5422	[68, 54, 13, 35]
K = 5	0.5643	[55, 46, 12, 35, 22]
K = 6	0.5110	[49, 41, 12, 32, 17, 19]
K = 7	0.5659	[30, 41, 10, 33, 17, 10, 29]
K = 8	0.5829	[30, 41, 10, 32, 14, 7, 15, 21]
K = 9	0.5503	[33, 24, 19, 32, 14, 7, 15, 21, 5]
K = 10	0.5723	[40, 32, 9, 13, 14, 7, 14, 10, 5, 26]

Table 1: Silhouette scores and cluster size distributions across different values of K.

As shown in Table 1, the global Silhouette Score peaks sharply at K = 8 with a value of 0.5829. Consequently, the model configured at K = 8 clusters was definitively selected as the structural baseline for further analyses.

With K = 8, we conducted an in-depth internal distribution analysis for each cluster to evaluate the convergence pattern of known longevity genes (Known = 1) versus candidate genes (Known = 0). The goal is to find high-density clusters of proven longevity genes, which will serve as a basis for inferring the function of candidate genes located within the same cluster.

Cluster	Total Genes	Known Longevity (Known=1)	Candidate (Known=0)	Known Gene Ratio (%)
1	30	10	20	33.3%
2	41	27	14	65.9%
3	10	5	5	50.0%
4	32	13	19	40.6%
5	14	8	6	57.1%
6	7	4	3	57.1%
7	15	7	8	46.7%
8	21	9	12	42.9%
TOTAL	170	83	87	48.8%

Table 2: Distribution and ratio of known longevity genes versus candidate genes across the K = 8 cluster topology

Statistical data with K = 8 in Table 2 shows a very clear differentiation in gene distribution between groups. The results showed that the genes were not spread out evenly among the groups. While the average percentage of known longevity genes across the entire dataset is 48.8%, meaning 83 out of 170 genes, Cluster 2 is completely dominant. It turned out to be the largest group with 41 genes in total, housing 27 known longevity genes and 14 candidate genes. Remarkably, 65.9% of the genes in this specific group are already proven to affect longevity, which is way higher than the dataset average.

This high concentration tells us that the genes in Cluster 2 work closely as a team. Because of this strong connection, we can say that the 14 candidate genes in Cluster 2 are our absolute best targets for future laboratory experiments to discover new life-extending genes.

Candidate Genes in Cluster 2	
YPL254W / HFI1	YDR264C / AKR1
YDR456W / NHX1	YMR161W / HLJ1
YBL007C / SLA1	YMR297W / PRC1
YOR209C / NPT1	YDR069C / DOA4
YOR023C / AHC1	YAL020C / ATS1
YBL047C / EDE1	YKR052C / MRS4
YKL205W / LOS1	YGR200C / ELP2

Table 3: List of potential candidate longevity genes identified within Cluster 2